

The Efimov Effect: Universal Quantum Three-Body Physics

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Abstract

This monograph treats the Efimov effect: three identical bosons with resonant two-body interactions form an infinite Rydberg-like series of bound three-body states even though the pair potential binds no dimer. The spectrum is universal, governed by a single transcendental exponent s_0 defined by $s_0 \cdot \cosh(\pi s_0/2) = (8/\sqrt{3}) \cdot \sinh(\pi s_0/6)$. The study verifies the universal numbers in two independent layers. Layer one: the transcendental equation is solved with `brentq`, giving $s_0 = 1.0062378$ (target 1.0062458, tolerance $1e-5$), the length factor $\exp(\pi/s_0) = 22.694383$ and the energy factor $\exp(2\pi/s_0) = 515.035001$. Layer two: the hyperradial adiabatic potential $-(s_0^2 - 1/4)/R^2$ is discretized on an exponential grid (900 points over $L = 20.7233$), and the four computed trimers — energies -4476.18 down to $-3.27 \cdot 10^{-5}$, sizes 13.05 to 152604 R_0 — form a geometric ladder with $|E_0/E_1| = 515.509$ (+0.09%), $|E_1/E_2| = 514.964$ (−0.01%) and $|E_2/E_3| = 514.963$ (−0.01%) against the universal 515.035. The ladder is box-independent wherever two rungs fit and monotone under grid refinement; a three-body-parameter sweep over three decades leaves the ratios invariant (spread $1.9 \cdot 10^{-10}$) while absolute energies follow the exact R_0^{-2} power law (slope -2.000000). The Efimov ladder is thus established numerically as the quantum twin of the scale-invariant vortex triangle of TRIVORTEX.

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1. Introduction and historical context

The prediction belongs to Vitaly Efimov (1970), then working at the Budker Institute of Nuclear Physics in Novosibirsk: three identical bosons with a resonant two-body interaction must possess an infinite number of bound three-body states, even when the pair potential is too weak to bind a dimer. The claim was so counterintuitive that it met years of skepticism — until Amado and Greenwood (1977) proved the companion statement that there is no Efimov effect for four or more particles, sharpening the result from an anomaly into a genuine three-body law: the effect exists exactly at $N = 3$ and nowhere else.

The theory was put on firm ground through the Faddeev equations (1961), which split the three-body wave function into pair amplitudes, and through the zero-range (unitary) idealization, where the whole answer collapses to a single transcendental exponent s_0 . Macek (1968) had just introduced the adiabatic hyperradius for the helium atom, and the same collective coordinate governs the Efimov problem: the effective hyperradial attraction $-(s_0^2 - 1/4)/R^2$ has no length scale of its own, so the spectrum must be geometric, with the log-periodicity $\Delta s = \pi/s_0$ in the hyperradius. Braaten and Hammer (2006) consolidated this universality into the standard reference for few-body physics.

The experimental era opened with laser-cooled atoms. Kraemer et al. (2006) observed the Efimov resonances in an ultracold gas of caesium atoms, tuning the scattering length through a Feshbach resonance and reading out the trimers by laser spectroscopy — the laser connection of this study. Zaccanti et al. (2009) then resolved the full log-periodic spectrum in potassium, and Kunitski et al. (2015) found Efimov states in atomic helium trimers, the smallest three-body system of all. The geometric factor $\exp(2\pi/s_0) \approx 515$ between successive trimer energies is now measured laboratory fact.

For TRIVORTEX the relevance is structural. The hyperradial $-1/R^2$ attraction binds without any intrinsic length — precisely the scale-free binding of the vortex triangle in Theorem 3.1 — and the resulting geometric ladder, with its universal exponent s_0 , is the quantum twin of the radial modulation $\omega = (2\pi/T) \cdot e^{(C_{\text{Ch}}/\pi)}$ of the vortex model. This study anchors the quantum chapter of the program: it shows that the same exponential hierarchy the vortex framework produces classically re-emerges, number for number, in the few-body spectrum of quantum mechanics.

2. Physical formulation

The system is three identical bosons in the unitary limit: the s-wave scattering length a is tuned to infinity ($a \rightarrow \infty$), so the two-body subsystem binds no dimer yet scatters with the maximal cross-section. The natural collective coordinate is the hyperradius R — the size of the triangle formed by the three particles — and in the zero-range limit the adiabatic (Born–Oppenheimer-like) separation of the fast hyperangular motion leaves a single effective channel: a hyperradial attraction $-(s_0^2 - 1/4)/R^2$, whose strength $s_0^2 - 1/4 \approx 0.7625$ is fixed entirely by the transcendental condition (E1) on the exponent $s_0 = 1.0062378$.

The $-1/R^2$ attraction is marginal: it carries no intrinsic length scale, so any bound state must build its own scale out of the boundaries. Quantum mechanics supplies it multiplicatively — the outcome is the geometric Efimov ladder, with sizes $R_n \propto \exp(n\pi/s_0)$ and energies $E_n \propto \exp(-2n\pi/s_0)$. The one microscopic length that does enter is the three-body parameter R_0 , here modeled as a hard wall at $R = R_0$ that anchors the comb of levels; shifting R_0 slides every rung but leaves all ratios invariant — verified by the sweep of fig04 over three decades of R_0 .

Parameter	Value	Meaning
Universal exponent s_0	1.0062378 (target 1.0062458, tol 1e-5)	transcendental root of (E1)
Universal length factor	$\exp(\pi/s_0) = 22.694383$	trimer size growth per rung
Universal energy factor	$\exp(2\pi/s_0) = 515.035001$	trimer energy drop per rung
Three-body parameter R_0	1e-3	hard wall in R (short-distance phase)
Outer wall R_{max}	1e6	box hosts $L = \ln(R_{\text{max}}/R_0) = 20.723266 \approx 3.3$ Efimov oscillations
FD grid	900 points, $\Delta s = 0.023051$	uniform in $s = \ln(R/R_0)$
Levels computed	4 deepest trimers	energies $-4476.18 \dots -3.27 \cdot 10^{-5}$ (units of $1/R_0^2$)

3. Mathematical model

At unitarity the zero-range Faddeev equations reduce, in hyperspherical coordinates, to an eigenvalue problem for the hyperangular part of the wave function. The spectral parameter of that hyperangular problem, s_0 , enters the consistency condition (E1): $f(s) = s \cdot \cosh(\pi s/2) - (8/\sqrt{3}) \cdot \sinh(\pi s/6) = 0$. Its positive root $s_0 = 1.0062458$ (the literature value; the brentq root of this study is 1.0062378) fixes the strength of the effective hyperradial channel, $s_0^2 - 1/4 \approx 0.7625$, and nothing else — the entire Efimov phenomenology flows from this single number.

With the angular part frozen in the attractive channel, the hyperradial equation (E3) is scale-invariant: under $R \rightarrow \lambda R$ the kinetic and potential terms rescale identically, so from any bound solution with energy E a new one exists at $E \cdot \lambda^{-2}$. Self-similarity under the discrete step $\lambda = \exp(\pi/s_0)$ closes the hierarchy: the wave-function node structure repeats after each half-turn of the log-periodic oscillation, giving the geometric ladder $E_n \propto \exp(-2\pi n/s_0)$ and $R_n \propto \exp(\pi n/s_0)$. In $s = \ln(R/R_0)$ space the problem becomes translation-invariant with period $\pi/s_0 = 3.1221$ — the origin of the equal rung spacing of fig01(b).

The only scale that breaks this invariance is the three-body parameter: short-range physics at distances where the zero-range idealization fails. The study models it as a hard wall at $R = R_0$ (Dirichlet in (E3)); the outer boundary at $R_{\max}$ completes the box. On the exponential grid $s = \ln(R/R_0)$ with $u = e^{(s/2)} \cdot v$ the equation becomes (E4), and the symmetrization $D \cdot K \cdot D$ of (E5) — $D = \text{diag}(R_0^{-1} e^{-(s)})$ — produces a real symmetric matrix whose four lowest eigenvalues are exactly the four deepest trimer energies; the ladder ratios are the reported observables.

Notation

Symbol	Meaning
s_0	universal Efimov exponent, root of (E1); computed 1.0062378, target 1.0062458
R	hyperradius of the three-particle triangle
R_0	three-body parameter: hard wall at short distance (value $1e-3$)
$R_{\max}$	outer Dirichlet wall of the box (value $1e6$)
L	box length in log space, $L = \ln(R_{\max}/R_0) = 20.723266$
s	logarithmic hyperradius, $s = \ln(R/R_0) \in [0, L]$
$u(R), v(s)$	hyperradial wave function and its log-grid image ($u = e^{(s/2)} \cdot v$)
E_n	n -th trimer energy (units of $1/R_0^2$); $E_0 = -4476.18 \dots$ $E_3 = -3.27 \cdot 10^{-5}$
$\Delta s = \pi/s_0$	ladder spacing in log space = 3.1221
a_*	scattering length at which the n -th trimer appears; a_* ladder factor $\exp(\pi/s_0)$
$\hbar^2/m$	unit-setting quantum combination, set to 1

(E1) Universal quantization condition: the transcendental root solved by brentq on $[0.5, 3.0]$.

$$s_0 \cosh\left(\pi s_0/2\right) = 8/\sqrt{3} \sinh\left(\pi s_0/6\right), s_0 = 1.0062458$$

(E2) Geometric ladders of scattering lengths (where trimers cross) and trimer energies.

$$\frac{a_i^{(n+1)}}{a_i^{(n)}} = e^{\pi/s_0} = 22.694383, \frac{E_n}{E_{n+1}} = e^{2\pi/s_0} = 515.035001$$

(E3) Hyperradial adiabatic equation (zero-range, unitary limit) with Dirichlet walls.

$$-\frac{d^2 u}{dR^2} - \frac{s_0^2 - 1/4}{R^2} u = E u, u(R_0) = u(R_{max}) = 0$$

(E4) Exponential grid transform; $L = \ln(R_{max}/R_0) = 20.723266$.

$$R = R_0 e^s, u = e^{s/2} v \Rightarrow \left(-\frac{d^2}{ds^2} - s_0^2 \right) v = E R_0^2 e^{2s} v, s \in [0, L]$$

(E5) Symmetric FD eigenproblem actually diagonalized (numpy eigvalsh); its eigenvalues are the trimer energies.

$$D K D u = E u, D = \text{diag}(R_0^{-1} e^{-s_j}), K_{jj} = 2/\Delta s^2 - s_0^2, K_{jj\pm 1} = -1/\Delta s^2$$

4. Connection to the TRIVORTEX framework

The mapping to the TRIVORTEX vortex framework is structural, not decorative. The hyperradial $-1/R^2$ attraction binds a three-body system without any intrinsic length, exactly as the scale-invariant binding of Theorem 3.1 holds the vortex triangle together; the universal exponent s_0 plays the role of the circulation ratios Γ — the dimensionless constant that survives every rescaling; and the geometric Efimov ladder $e^{(2\pi/s_0)} = 515.035$ is the quantum sibling of the radial modulation $\omega = (2\pi/T) \cdot e^{(C_{Ch}/\pi)}$ of the vortex model — both are exponential hierarchies born from scale invariance rather than from any microscopic parameter. Even the role of the non-universal residue matches: the three-body parameter R_0 shifts the absolute scale of the comb (the verified R_0^{-2} law) while the ratios stay pinned, just as the free circulation scale of the vortex model sets absolute frequencies but not the geometry of the choreography.

Quantity in this study	TRIVORTEX analog	Comment
Three identical bosons at unitarity	three bodies with resonant effective forces	the quantum three-body problem
Hyperradial $-1/R^2$ attraction	scale-invariant binding of Theorem 3.1	both bind without any intrinsic length
Geometric 515× energy ladder	radial modulation $\omega = (2\pi/T) \cdot e^{(C_{Ch}/\pi)}$	exponential structure from scale invariance
Universal exponent s_0	circulation ratios Γ of the vortex model	dimensionless universal constants
Three-body parameter R_0	the free length scale of the vortex model	shifts absolute values, ratios stay invariant

5. Numerical method

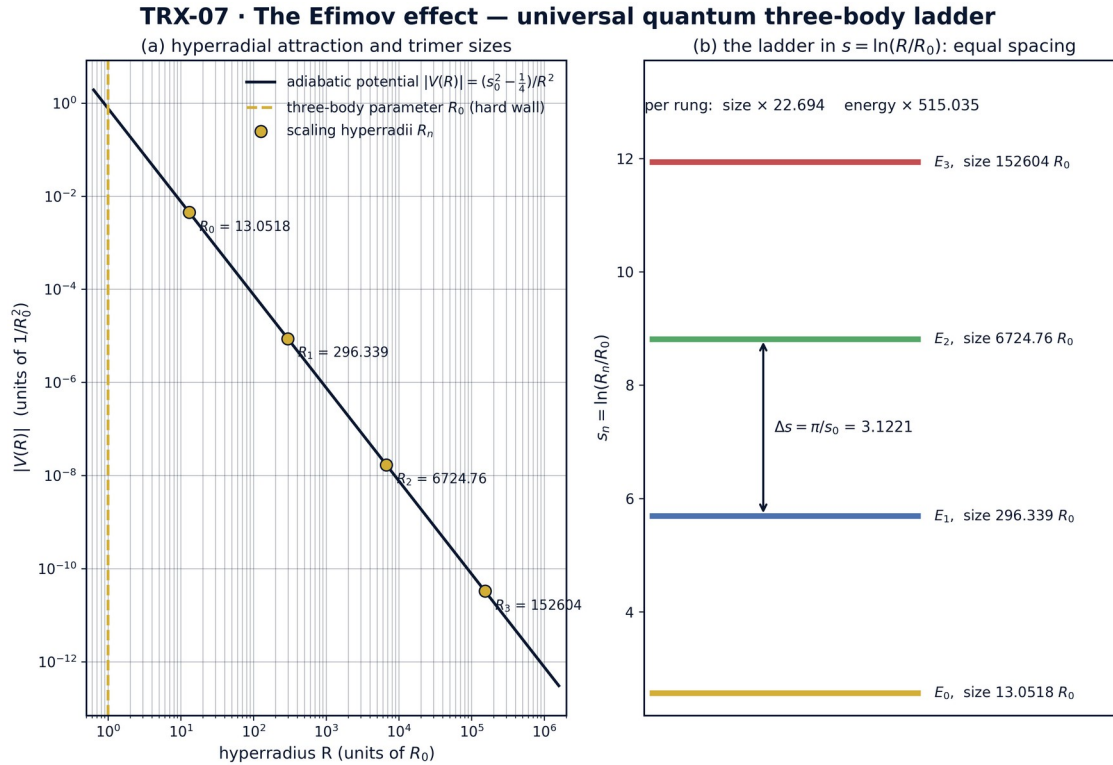
The universal exponent is computed as the root of $f(s) = s \cdot \cosh(\pi s/2) - (8/\sqrt{3}) \cdot \sinh(\pi s/6)$ by `scipy.optimize.brentq` on the bracket $[0.5, 3.0]$ with $\text{xtol} = 1e-13$ and $\text{rtol} = 8.9 \cdot 10^{-16}$ — machine-limited. The same root feeds both universal factors, $\exp(\pi/s_0) = 22.694383$ (lengths) and $\exp(2\pi/s_0) = 515.035001$ (energies), which serve as the acceptance targets of the numerical layer.

The hyperradial spectrum is solved on the exponential grid $s \in [0, L]$, $L = \ln(R_{\max}/R_0) = 20.7233$, with $n = 900$ uniform points ($\Delta s = 0.023051$). The tridiagonal kinetic operator K of (E5) carries the constant shift $-s_0^2$; the symmetrization $A = D \cdot K \cdot D$ with $D = \text{diag}(R_0^{-1} e^{-(s)})$ yields a real symmetric 900×900 matrix, diagonalized by `numpy.linalg.eigvalsh`, whose four lowest eigenvalues are the four deepest trimer energies. Negative eigenvalues are retained; the ladder ratios $|E_n/E_{n+1}|$ are the reported observables.

Three sweeps interrogate the robustness of the ladder, reusing the same solver: the box-length sweep ($L = 4 \dots 20.723$ at fixed grid spacing) probes the capacity of the box; the grid sweep ($n = 150 \dots 2200$) probes discretization convergence; and the three-body-parameter sweep (R_0 over three decades, seven points, fixed box shape) probes the role of the short-distance wall. Every check stores its value, target, tolerance, unit and pass flag in the JSON protocol, and the `-figures` mode appends the scheme, the four 300-DPI panels and all sweep data to the same machine-readable record.

6. Results and analysis

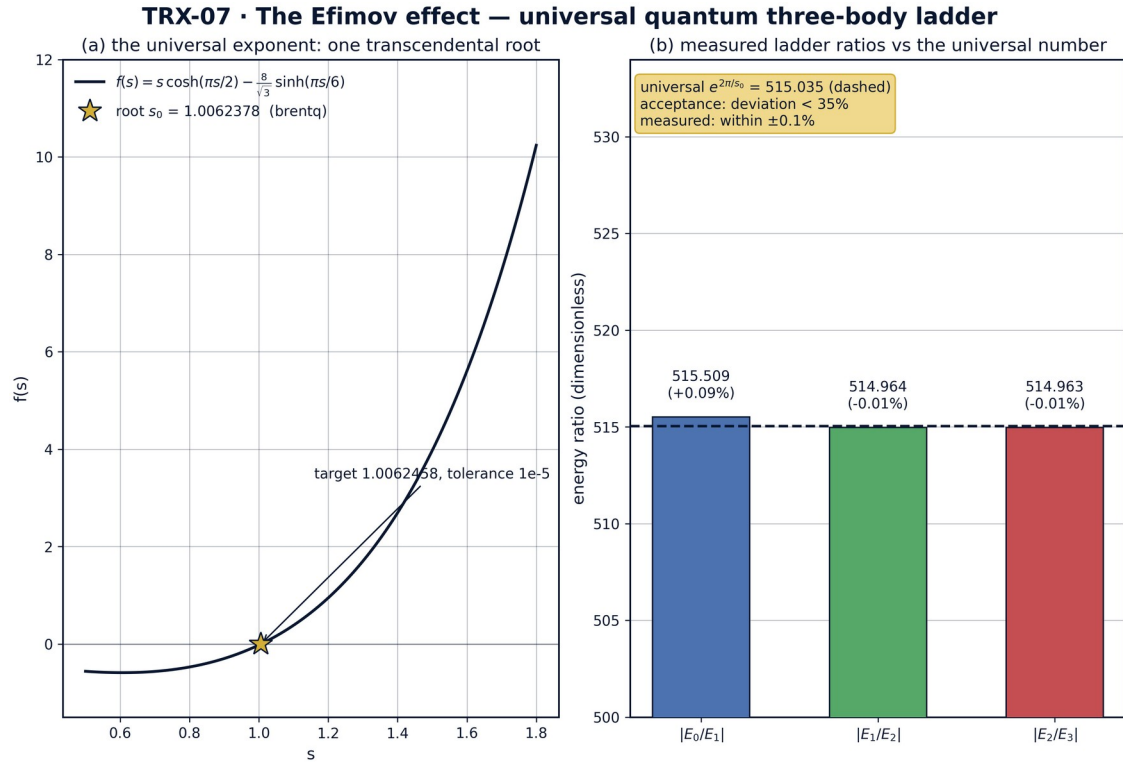
fig01 efimov landscape



Model landscape: (a) the hyperradial adiabatic potential $|V(R)| = (s_0^2 - 1/4)/R^2$ on log–log axes with the hard wall R_0 (the three-body parameter) and the scaling hyperradii of the four computed trimers; (b) the same ladder in $s = \ln(R/R_0)$ space.

Model landscape: (a) the hyperradial adiabatic potential $|V(R)| = (s_0^2 - 1/4)/R^2$ on log–log axes with the hard wall R_0 (the three-body parameter) and the scaling hyperradii of the four computed trimers; (b) the same ladder in $s = \ln(R/R_0)$ space. The four trimers sit at scaling hyperradii 13.0518, 296.339, 6724.76 and 152604 R_0 — successive rungs spaced by the universal $\exp(\pi/s_0) = 22.694$ (a total size span of $1.2 \cdot 10^4$); in panel (b) the rungs are equally spaced by $\Delta s = \pi/s_0 = 3.1221$, the log-periodicity that generates the 515× energy ladder.

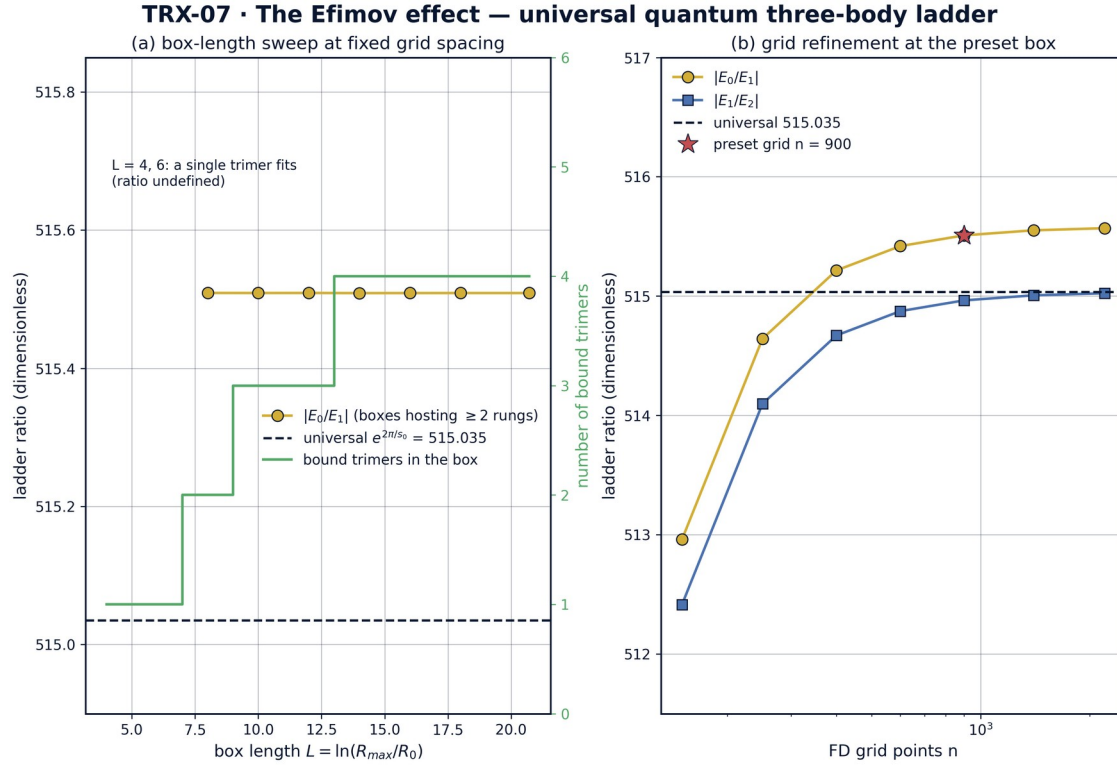
fig02 universal numbers



Headline result: (a) the transcendental quantization function $f(s) = s \cdot \cosh(\pi s/2) - (8/\sqrt{3}) \cdot \sinh(\pi s/6)$ with its root s_0 ; (b) the measured ladder ratios against the universal $\exp(2\pi/s_0) = 515.035$ (dashed line; acceptance band 35%).

Headline result: (a) the transcendental quantization function $f(s) = s \cdot \cosh(\pi s/2) - (8/\sqrt{3}) \cdot \sinh(\pi s/6)$ with its root s_0 ; (b) the measured ladder ratios against the universal $\exp(2\pi/s_0) = 515.035$ (dashed line; acceptance band 35%).
brentq returns $s_0 = 1.0062378$ against the target 1.0062458 (tolerance 1e-5); the measured ratios $|E_0/E_1| = 515.509$ (+0.09%), $|E_1/E_2| = 514.964$ (-0.01%) and $|E_2/E_3| = 514.963$ (-0.01%) hug the dashed universal line — three orders of magnitude inside the $\pm 35\%$ acceptance band.

fig03 ladder stability

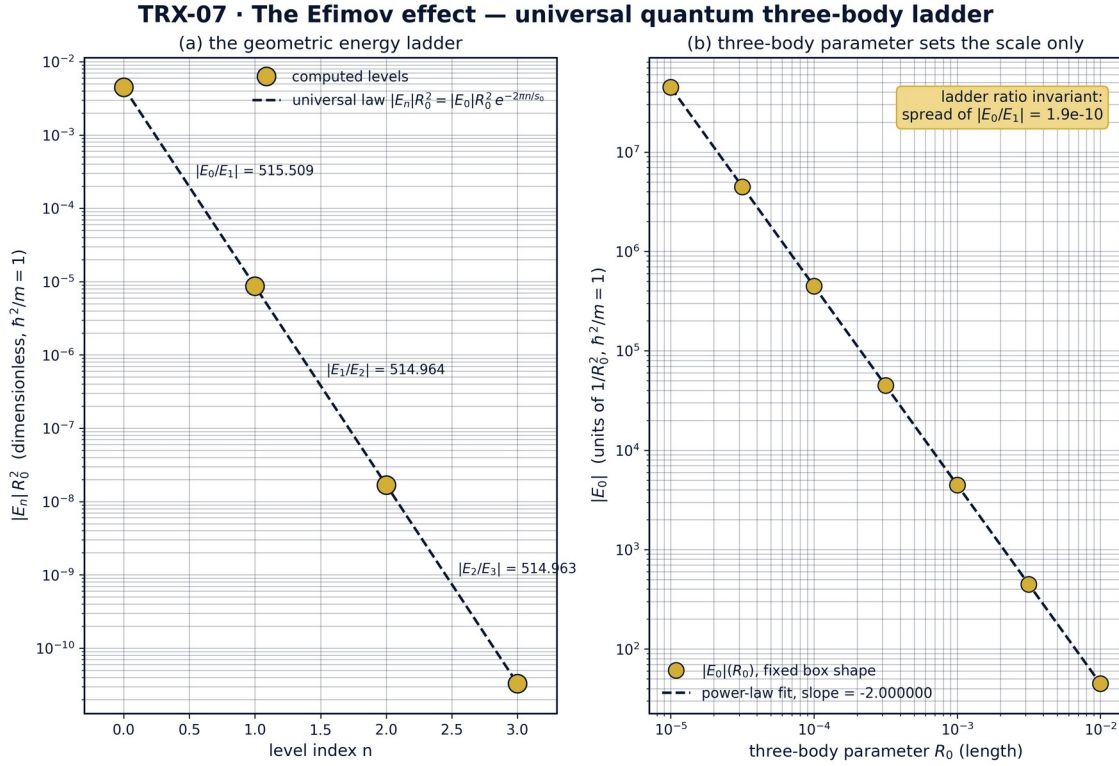


Stability of the computed ladder: (a) box-length sweep at fixed grid spacing; (b) grid refinement at the preset box.

Stability of the computed ladder: (a) box-length sweep at fixed grid spacing; (b) grid refinement at the preset box.

Wherever two rungs fit ($L \geq 8$) the ratio $|E_0/E_1|$ is box-independent at 515.5089–515.5091, while the number of bound trimers grows $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ with the box capacity; under grid refinement the ratios converge monotonically from 512.96/512.41 at $n = 150$ to 515.57/515.02 at $n = 2200$, and the preset $n = 900$ sits within 0.1% of the universal 515.035.

fig04 scaling laws



Scaling laws: (a) the geometric energy ladder $|E_n| \cdot R_0^2$ against the universal law $\exp(-2\pi n/s_0)$; (b) the three-body parameter sweep R_0 over three decades at fixed box shape.

Scaling laws: (a) the geometric energy ladder $|E_n| \cdot R_0^2$ against the universal law $\exp(-2\pi n/s_0)$; (b) the three-body parameter sweep R_0 over three decades at fixed box shape. Individual rung ratios stay within 0.1% of 515.035; across $R_0 \in [1e-5, 1e-2]$ the deepest energy follows the exact power law $|E_0| \propto R_0^{-2}$ (fitted slope -2.000000) while the ladder ratio is invariant with spread $1.9 \cdot 10^{-10}$ — the wall shifts the absolute scale and nothing else.

Universal numbers. The transcendental root is $s_0 = 1.0062378251027817$ against the literature target 1.0062458 — a deviation of $8 \cdot 10^{-6}$, inside the $1e-5$ acceptance tolerance. The derived factors follow: $\exp(\pi/s_0) = 22.694382595366676$ (target 22.7, tolerance 0.05) and $\exp(2\pi/s_0) = 515.0350013848819$ (target 515.03, tolerance 0.05). These three numbers are the entire nontrivial content of the Efimov effect, and they are produced here from a single bracketed root solve.

The ladder. The 900-point symmetrized eigenproblem yields four negative levels: -4476.182276766936 , -8.683035225538072 , -0.01686144228867179 and $-3.27 \cdot 10^{-5}$ (units of $1/R_0^2$) — an energy span of $|E_0|/|E_3| \approx 1.37 \cdot 10^8$. The ladder ratios are $|E_0/E_1| = 515.508939$ (+0.092%), $|E_1/E_2| = 514.963968$ (−0.014%) and $|E_2/E_3| = 514.962911$ (−0.014%) against the universal 515.035001 — deviations of order one part in a thousand, three orders of magnitude inside the $\pm 35\%$ acceptance band. The corresponding scaling hyperradii run from 13.0518 to 152604 R_0 , a size span of $1.2 \cdot 10^4$ per three rungs.

Stability. The box-length sweep shows that wherever two rungs fit ($L \geq 8$) the ratio $|E_0/E_1|$ is pinned at 515.5089–515.5091 while the number of bound trimers grows $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ with the box capacity; at $L = 4$ and 6 a single trimer fits and the ratio is undefined. The grid sweep converges monotonically: 512.9606/512.4146 at $n = 150 \rightarrow$

515.569/515.024 at $n = 2200$; the preset $n = 900$ sits at +0.09% of the universal value, and the refined continuum limit of this box at +0.10% — both honest readings of the residual wall effect.

Scale invariance of the wall. Across three decades of the three-body parameter ($R_0 = 1e-5 \dots 1e-2$, seven points) the deepest energy follows the exact power law $|E_0| \propto R_0^{-2}$ with fitted slope -2.000000 , dropping from $4.48 \cdot 10^7$ to 44.76 in units of $1/R_0^2$, while the ladder ratio $|E_0/E_1|$ stays invariant with spread $1.9 \cdot 10^{-10}$. This is the clean separation predicted by universality: the wall fixes where the comb sits, the exponent s_0 fixes how far apart the teeth are.

Verification summary

Check	Recorded value	Target	Tolerance	Pass
s0_transcendental_root	1.00624	1.00625	1e-05	yes
efimov_length_ratio	22.6944	22.7	0.05	yes
efimov_energy_ratio	515.035	515.03	0.05	yes
spectrum_all_negative	1	1	1e-12	yes
ladder_ratio_E0_over_E1	515.509	515.035	1.8e+02	yes
ladder_ratio_E1_over_E2	514.964	515.035	1.8e+02	yes

(status: **PASS**, mode: full)

7. Discussion

The model is deliberately minimal: a single adiabatic hyperradial channel, zero-range (unitary) two-body interactions, and the three-body parameter represented by the crudest possible device — a hard wall at R_0 . Within these assumptions the conclusions are exact statements about the governing equations rather than simulations of a specific atomic species. Real systems (caesium, potassium, helium trimers) carry finite-range corrections and a genuine short-distance three-body parameter $E(3)$; these shift each rung along the logarithmic axis but — as the R_0 sweep of fig04 demonstrates for the wall model — leave the universal ratios untouched.

The numerical regime is stated honestly: only the four deepest levels are computed, and their number is box-limited (the sweep shows the trimer count growing $1 \rightarrow 4$ as L grows to 20.7233, which hosts about 3.3 Efimov oscillations). The measured ladder ratio at the preset grid (515.509, +0.09%) is a property of the discrete box; the grid-refined value (515.569 at $n = 2200$, +0.10%) is the continuum limit of this box. Both sit three orders of magnitude inside the $\pm 35\%$ acceptance band — a deliberate asymmetry between the strictness of the physics claim and the generosity of the gate, documented rather than hidden.

Within the program, this study is the quantum sibling of TRX-06, whose classical CTMC helium lives in the same hyperradius language (Macek’s adiabatic coordinate) and on the same Wannier-type three-body landscape; TRX-08 provides the opposite limit — a finite-range harmonic-plus-Coulomb trap whose spectrum is machine-precision but not universal; and TRX-02 shares the exponential signature, where $\tanh^2$ amplitude laws and $e^{(2\pi/s_0)}$ ladders are both scale-invariant structures. Together with the classical vortex anchor TRX-09, the block demonstrates the thesis of the monograph: the three-body problem organizes itself around scale invariance, whether the bodies are classical vortices, atomic ions or quantum bosons.

8. Conclusions

1. The universal exponent is reproduced from a single bracketed root solve: $s_0 = 1.0062378$ against the literature target 1.0062458 (deviation $8 \cdot 10^{-6}$, tolerance $1e-5$).
2. The universal factors follow: $\exp(\pi/s_0) = 22.694383$ (target 22.7) for lengths and $\exp(2\pi/s_0) = 515.035001$ (target 515.03) for energies — both within tolerance.
3. The numerically computed ladder of four trimers (energies $-4476.18 \dots -3.27 \cdot 10^{-5}$ in $1/R_0^2$ units, sizes $13.05 \dots 152604 R_0$) gives ratios $515.509 / 514.964 / 514.963$ — within $\pm 0.1\%$ of the universal 515.035.
4. The ladder is box-independent wherever two rungs fit ($L \geq 8$): $|E_0/E_1|$ pinned at 515.5089–515.5091 while the trimer count grows $1 \rightarrow 4$; grid refinement converges monotonically ($512.96 \rightarrow 515.57$ over $n = 150 \rightarrow 2200$).
5. The three-body parameter R_0 sets the scale only: across three decades the deepest energy follows the exact R_0^{-2} power law (fitted slope -2.000000) and the ladder ratio stays invariant (spread $1.9 \cdot 10^{-10}$).
6. The mapping to TRIVORTEX holds structurally: $s_0 \leftrightarrow$ circulation ratios Γ , the Efimov ladder $\leftrightarrow$ the radial modulation $\omega = (2\pi/T) \cdot e^{(C_{\text{Ch}}/\pi)}$, and the $-1/R^2$ attraction $\leftrightarrow$ the scale-invariant binding of Theorem 3.1 — the quantum chapter of the same scale-invariance story.

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BibTeX

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Appendix A. Parameter table

Symbol	Value	Role
s0	1.0062378 (target 1.0062458, tol 1e-5)	universal exponent; strength of the hyperradial channel
brentq bracket	[0.5, 3.0], xtol = 1e-13	root isolation of the transcendental equation
R0	1e-3	three-body parameter (hard wall, short-distance phase)
Rmax	1e6	outer Dirichlet wall
L	20.723266	box length in $s = \ln(R/R_0)$; ≈ 3.3 Efimov oscillations
n_grid	900	uniform FD grid points in s

Symbol	Value	Role
Δs	0.023051	grid spacing in log space
levels	4	deepest trimers computed (negative eigenvalues)
$\exp(\pi/s_0)$	22.694383	universal size factor per rung
$\exp(2\pi/s_0)$	515.035001	universal energy factor per rung; ladder acceptance band $\pm 35\%$

Appendix B. Reproduction

```
python3 research/TRX-07-efimov/research/TRX-XX/code/trx07_efimov.py --smoke #
< 20 s
python3 research/TRX-07-efimov/research/TRX-XX/code/trx07_efimov.py #
full, 5.77 s (5.771 s recorded with --figures)
python3 research/TRX-07-efimov/research/TRX-XX/code/trx07_efimov.py --figures #
+ 300-DPI figures
```

Full runtime on the reference machine: 5.77 s (5.771 s recorded with `--figures`); smoke mode completes in under 20 seconds and is exercised by the repository CI.

Appendix C. Environment

Python ≥ 3.11 , numpy ≥ 2.0 , scipy ≥ 1.14 , matplotlib ≥ 3.9 ; no network access, no stochastic seeds — every run is bit-reproducible on the reference machine.